

Roshan Hegde

 github.com/RoshanAH  rhegde@illinois.edu  (952) 594-4175

EDUCATION

University of Illinois at Urbana-Champaign

B.S. in Computer Science, Grainger College of Engineering

B.S. in Mathematics, College of Liberal Arts & Sciences

May 2026

GPA: 3.91/4.0

Highland Park High School

May 2023

GPA: 4.69/4.0

RELEVANT COURSEWORK

Courses: Advanced Algorithms, Operating Systems Design, Computer Architecture, Programming Languages and Compilers, Cryptography, Data Structures (Honors), Probability and Statistics for Computer Science, Linear Algebra with Computational Applications

SKILLS

Languages: C, C++, Go, Java, Kotlin, Python, Lua, Rust, Typescript

Tools: Git, GitHub, Docker, Linux, Nix, Vim

Frameworks: Svelte, React, Tailwindcss, Vite, AWS

WORK EXPERIENCE

Olam Food and Ingredients, Chicago IL | *Infrastructure Development Intern* | *Azure, python* May – August 2025

- Analyzed current utilization of cloud assets and identified opportunities to improve cost efficiency by 12%
- Developed a data analysis service to automate VM performance reporting to identify over-allocated resources

MechMania, Champaign IL | *Lead Infrastructure Developer* | *Go, AWS, python* August 2024 – Present

- Built a multithreaded web-socket server in Go interacting with AWS services including databases and cloud storage
- Implemented a Java-Python language interop system for broader audience reach

AWARDS

1st place Developer Tool, Hackillinois '25 | *Rust, Cargo, Git* February 2024

- Competed with over 1,100 other students at Illinois' largest Hackathon
- Developed a full stack developer tool intended for rapid prototyping of data pipelines
- Defined a strongly typed JSON based protocol for ease of extensibility and versatility

1st place, MechMania '24 | *Java, Python, Git* September 2024

- Competed in a 24-hour Game AI Hackathon held annually since 1995 where students' AI models compete
- Fine-tuned heuristics to pilot dogfighting planes, leveraging predictive modeling to gain a competitive edge

1st place, Hackillinois '24 | *Rust, Cargo, Git* February 2024

- Built a developer tool used to universally initialize a basic development environment for over 30 languages
- Designed modular software architecture allowing for native configuration and rapid extensibility

1st place, MechMania '23 | *Java, Python, Git* September 2023

- Implemented A* pathfinding with a modified Manhattan distance heuristic
- Employed a divide and conquer strategy in tandem with pathfinding to outperform other models

PROJECTS

nix-tinker | *Rust, Cargo, Nix, Nix Flakes*

May 2024 – Present

- Developed an open-source tool for rapid prototyping of Nix configuration files, streamlining development workflow
- Crafted an intuitive CLI for managing file links with the Nix store, significantly enhancing user efficiency
- Leveraged Nix Flakes to guarantee reproducible and consistent build environments across diverse systems